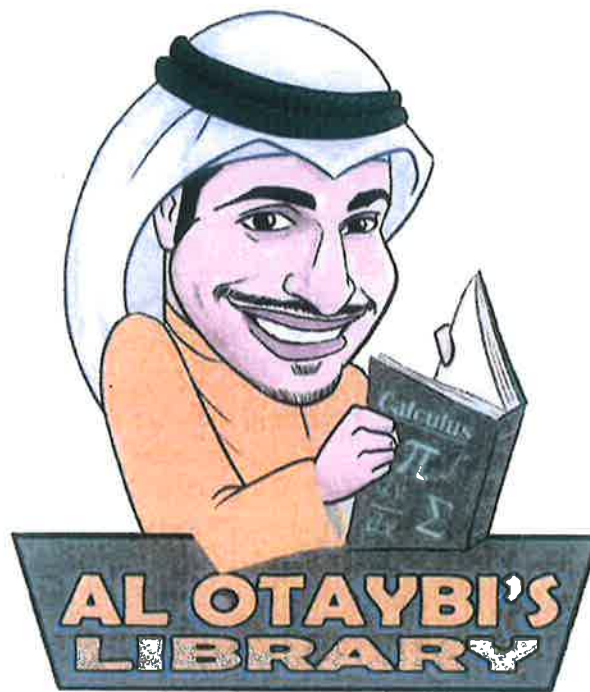


Kuwait University

2021/2022

Spring course



Calculus B

First

Section : 7.3

7.3 Trigonometric Substitution.

(5)



Calculus B

Quiz # 4



$$\int \cos(5x) dx$$

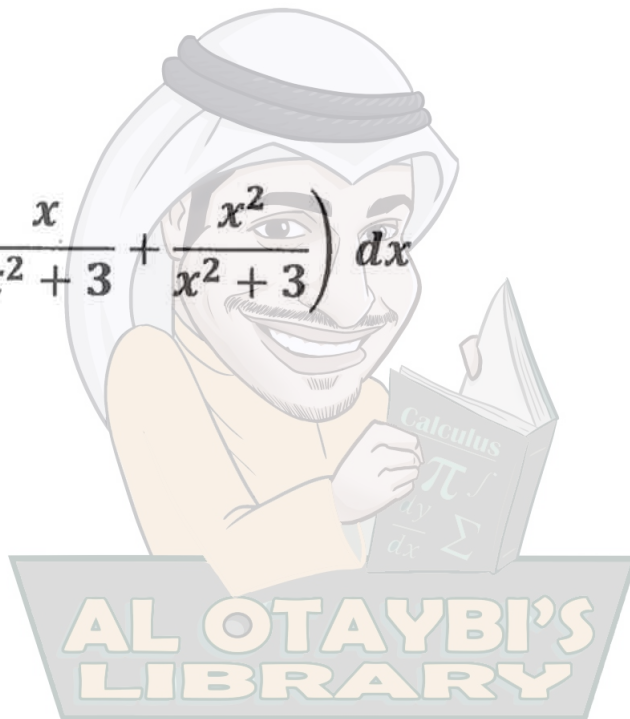
$$\int e^{3x} dx$$

$$\int \left(\frac{1}{\sqrt{2-x^2}} + \frac{1}{x\sqrt{x^2-2}} + \frac{x}{\sqrt{x^2-2}} \right) dx$$

$$\int \left(\frac{1}{x^2+3} + \frac{x}{x^2+3} + \frac{x^2}{x^2+3} \right) dx$$

$$\int x^3 3^{7x^2} dx$$

$$\int \frac{\cos^2 x \sqrt{\sin x}}{\tan x} dx$$



أنواع التكاملات



By parts

عند وجود أولوية من الأولويات



u

dv

بـس دوال مثلثية

أو (h)

حدودية من الدرجة الثانية

حدودية من الدرجة الثانية
($\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$)



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خطوات الحل



(1) الفرض بحسب الحالة

$$\sqrt{a^2 - x^2}$$



$$x = a \sin \theta$$

$$\sqrt{a^2 + x^2}$$



$$x = a \tan \theta$$

$$\sqrt{x^2 - a^2}$$



$$x = a \sec \theta$$

أرسم مثلث

أشتق للحصول على ما يعادل
 dx

(2) من الفرض

(3) أشيل وأبدل , ثم أحل السؤال بدرس الدوال المثلثية

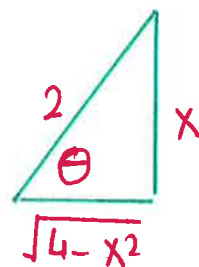
المثلث

الفرض

(4) أرد كل الناتج بصيغته الأصلية عن طريق

$$\int \frac{x^2}{\sqrt{4-x^2}} dx$$

Let $x = 2 \sin \theta \longrightarrow \frac{x}{2} = \sin \theta$
 $dx = 2 \cos \theta d\theta$



$$\int \frac{(2 \sin \theta)^2}{\sqrt{4 - (2 \sin \theta)^2}} \cdot 2 \cos \theta d\theta$$

$$\begin{aligned} \sqrt{4 - 4 \sin^2 \theta} &= \sqrt{4(1 - \sin^2 \theta)} = \sqrt{4 \cos^2 \theta} \\ &= 2 \cos \theta \end{aligned}$$

$$\int \frac{4 \sin^2 \theta}{2 \cos \theta} \cdot 2 \cos \theta d\theta$$

$$4 \int \sin^2 \theta d\theta$$

$$4 \cdot \frac{1}{2} \int (1 - \cos 2\theta) d\theta$$

$$2 \left[\theta - \frac{\sin 2\theta}{2} \right] + C$$

$$2 \left[\theta - \frac{2 \sin \theta \cos \theta}{2} \right] + C$$

$$2 \left[\sin^{-1} \left(\frac{x}{2} \right) - \frac{x}{2} \cdot \frac{\sqrt{4-x^2}}{2} \right] + C$$

لازم أرد كل شي الى x

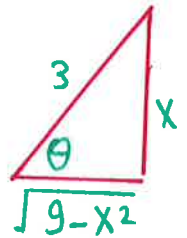
عن طريق

الفرض

مثلت

$$\int \frac{1}{x^2 \sqrt{9-x^2}} dx$$

Let $x = 3 \sin \theta \longrightarrow \frac{x}{3} = \sin \theta$
 $dx = 3 \cos \theta d\theta$



$$\int \frac{1}{(3 \sin \theta)^2 \sqrt{9 - (3 \sin \theta)^2}} 3 \cos \theta d\theta$$

$$\begin{aligned} \sqrt{9 - 9 \sin^2 \theta} &= \sqrt{9(1 - \sin^2 \theta)} \\ &= \sqrt{9 \cos^2 \theta} = 3 \cos \theta \end{aligned}$$

$$\int \frac{1}{9 \sin^2 \theta} \cancel{3 \cos \theta} \cancel{3 \cos \theta} d\theta$$

$$\frac{1}{9} \int \frac{1}{\sin^2 \theta} d\theta$$

$$\frac{1}{9} \int \csc^2 \theta d\theta$$

$$-\frac{1}{9} \cot \theta + C$$

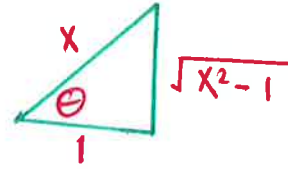
$$-\frac{1}{9} \frac{\sqrt{9-x^2}}{x} + C$$

أي دالة مثلثية بروحها
بالمقام تصعد لمقلوبها

$$\int \frac{x^3}{\sqrt{x^2 - 1}} dx$$

Let $x = \sec \theta$

$dx = \sec \theta \tan \theta d\theta$



$$\int \frac{\sec^3 \theta}{\sqrt{\sec^2 \theta - 1}} \sec \theta \tan \theta d\theta$$

$\sqrt{\tan^2 \theta} = \tan \theta$

$$\int \frac{\sec^3 \theta}{\tan \theta} \sec \theta \tan \theta d\theta$$

$$\int \sec^4 \theta d\theta$$

زوي

$$\int \sec^2 \theta \sec^2 \theta d\theta$$

$$\int (1 + \tan^2 \theta) \sec^2 \theta d\theta$$

Let $u = \tan \theta$

$du = \sec^2 \theta d\theta$

$$\int (1 + u^2) du = u + \frac{u^3}{3} + C$$

$$= \tan \theta + \frac{(\tan \theta)^3}{3} + C$$

$$= \frac{\sqrt{x^2 - 1}}{1} + \frac{\left(\frac{\sqrt{x^2 - 1}}{1}\right)^3}{3} + C$$

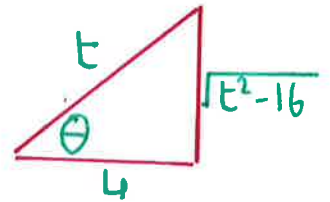


Text Book Question



$$\int \frac{dt}{t^2 \sqrt{t^2 - 16}}$$

$$\text{Let } t = 4 \sec \theta \longrightarrow \frac{t}{4} = \sec \theta$$
$$dt = 4 \sec \theta \tan \theta d\theta$$



$$\int \frac{4 \sec \theta \tan \theta}{(4 \sec \theta)^2 \sqrt{(4 \sec \theta)^2 - 16}} d\theta$$

$$\int \frac{4 \sec \theta \tan \theta}{16 \sec^2 \theta \sqrt{16 \sec^2 \theta - 16}} d\theta$$
$$\sqrt{16 \sec^2 \theta - 16} = \sqrt{16(\sec^2 \theta - 1)} = \sqrt{16 \tan^2 \theta} = 4 \tan \theta$$

$$\frac{1}{16} \int \frac{1}{\sec \theta} d\theta$$

$$\frac{1}{16} \int \cos \theta d\theta$$

$$\frac{1}{16} \sin \theta + C$$

$$\frac{1}{16} \frac{\sqrt{t^2 - 16}}{t} + C$$

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LIBRARY

$$\int \frac{1}{x^2 (x^2 + 1)^{\frac{3}{2}}} dx$$

Let $x = \tan \theta$

$dx = \sec^2 \theta d\theta$



$$\int \frac{1}{\tan^2 \theta (\tan^2 \theta + 1)^{\frac{3}{2}}} \cdot \sec^2 \theta d\theta$$

$$\int \frac{1}{\tan^2 \theta \sec^3 \theta} \sec^2 \theta d\theta$$

$$\int \frac{1}{\tan^2 \theta \sec \theta} d\theta = \int \frac{1}{\frac{\sin^2 \theta}{\cos^2 \theta} \frac{1}{\cos \theta}} d\theta$$

$$\int \frac{\cos^3 \theta}{\sin^2 \theta} d\theta = \int \cos^2 \theta (\sin \theta)^{-2} d\theta$$

$$\int \cos^2 \theta (\sin \theta)^{-2} \cos \theta d\theta$$

$$\int (1 - \sin^2 \theta) (\sin \theta)^{-2} \cos \theta d\theta$$

$u = \sin \theta$

$du = \cos \theta d\theta$

$$\rightarrow \int (1 - u^2) u^{-2} du$$

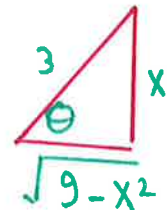
$$\int (u^{-2} - 1) du = \frac{u^{-1}}{-1} - u + C = -\frac{1}{u} - u + C$$

$$-\frac{1}{\sin \theta} - \sin \theta + C$$

$$-\frac{\sqrt{x^2 + 1}}{x} - \frac{x}{\sqrt{x^2 + 1}} + C$$

$$\int \frac{x^2}{(9-x^2)^{\frac{3}{2}}} dx$$

Let $x = 3\sin\theta \longrightarrow \frac{x}{3} = \sin\theta$
 $dx = 3\cos\theta d\theta$



$$\int \frac{(3\sin\theta)^2}{[9-(3\sin\theta)^2]^{\frac{3}{2}}} 3\cos\theta d\theta$$

$$\begin{aligned} &\longrightarrow [9-9\sin^2\theta]^{\frac{3}{2}} = [9(1-\sin^2\theta)]^{\frac{3}{2}} \\ &= [9\cos^2\theta]^{\frac{3}{2}} = (3\cos\theta)^3 = 27\cos^3\theta \end{aligned}$$

$$\int \frac{\cancel{9}\sin^2\theta}{\cancel{27}\cos^3\theta} \cancel{3\cos\theta} d\theta$$

$$\int \frac{\sin^2\theta}{\cos^2\theta} d\theta = \int \tan^2\theta d\theta$$

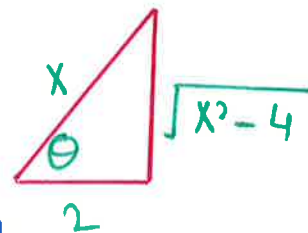
$$\int (\sec^2\theta - 1) d\theta$$

$$= \tan\theta - \theta + C$$

$$= \frac{x}{\sqrt{9-x^2}} - \sin^{-1}\left(\frac{x}{3}\right) + C$$

$$\int \frac{1}{x^3 \sqrt{x^2 - 4}} dx$$

Let $x = 2 \sec \theta \longrightarrow \frac{x}{2} = \sec \theta$
 $dx = 2 \sec \theta \tan \theta d\theta$



$$\int \frac{1}{(2 \sec \theta)^3 \sqrt{(2 \sec \theta)^2 - 4}} \cdot 2 \sec \theta \tan \theta d\theta$$

$$\sqrt{4 \sec^2 \theta - 4} = \sqrt{4(\sec^2 \theta - 1)} = \sqrt{4 \tan^2 \theta} = 2 \tan \theta$$

$$\int \frac{1}{8 \sec^3 \theta} \cdot \cancel{2 \sec \theta} \cancel{\tan \theta} d\theta$$

$$\int \frac{1}{8 \sec^2 \theta} d\theta = \frac{1}{8} \int \cos^2 \theta d\theta$$

$$\frac{1}{8} \cdot \frac{1}{2} \int (1 + \cos 2\theta) d\theta$$

$$\frac{1}{16} \left[\theta + \frac{\sin 2\theta}{2} \right] + C$$

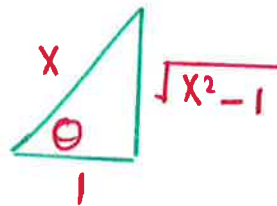
$$\frac{1}{16} \left[\theta + \frac{2 \sin \theta \cos \theta}{2} \right] + C$$

$$\frac{1}{16} \left[\sec^{-1} \left(\frac{x}{2} \right) + \frac{\sqrt{x^2 - 4}}{x} \cdot \frac{2}{x} \right] + C$$

$$\int \frac{x^6 + 2}{x^3 \sqrt{x^2 - 1}} dx$$

Let $x = \sec \theta$

$$dx = \sec \theta \tan \theta d\theta$$



$$\int \frac{\sec^6 \theta + 2}{\sec^3 \theta \sqrt{\sec^2 \theta - 1}} \cdot \sec \theta \tan \theta d\theta$$

$$\int \frac{\sec^6 \theta + 2}{\sec^2 \theta \tan \theta} \cdot \sec \theta \tan \theta d\theta$$

$\sqrt{\tan^2 \theta} = \tan \theta$

$$\int \frac{\sec^6 \theta + 2}{\sec^2 \theta} d\theta = \int \left[\frac{\sec^6 \theta}{\sec^2 \theta} + \frac{2}{\sec^2 \theta} \right] d\theta$$

$$\int \sec^4 \theta d\theta + \int 2 \cos^2 \theta d\theta$$

زنجی

$$\int \sec^2 \theta \sec^2 \theta d\theta \quad 2 \cdot \frac{1}{2} \int (1 + \cos 2\theta) d\theta$$

$$\int (1 + \tan^2 \theta) \sec^2 \theta d\theta$$

$$u = \tan \theta$$

$$du = \sec^2 \theta d\theta$$

$$\int (1 + u^2) du = u + \frac{u^3}{3} + C_1$$

$$= \tan \theta + \frac{(\tan \theta)^3}{3} + C_1$$

$$\theta + \frac{\sin 2\theta}{2} + C_2$$

$$\theta + \frac{2 \sin \theta \cos \theta}{2} + C_2$$

$$\Rightarrow \frac{\sqrt{x^2 - 1}}{1} \cdot \frac{\left(\frac{\sqrt{x^2 - 1}}{1}\right)^3}{3} + \sec^{-1}(x) + \frac{\sqrt{x^2 - 1}}{x} \cdot \frac{1}{x} + C$$



الفكرة الثانية



Completing square

عند وجود

x^2 مع x

(1) جعل معامل x^2 = 1

(2) تطبيق قاعدة السهم

Examples

$$8x - x^2$$

$$- [x^2 - 8x]$$

$$- [(x - 4)^2 - 16]$$



$$16 - (x - 4)^2$$

$$x^2 + 6x - 7$$

$$(x + 3)^2 - 9 - 7$$



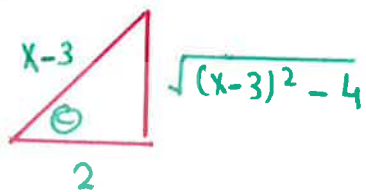
$$(x + 3)^2 - 16$$

$$\int \frac{2x-1}{\sqrt{x^2-6x+5}} dx$$

$$\int \frac{2x-1}{\sqrt{(x-3)^2-4}} dx$$

$$\begin{aligned} x^2-6x+5 \\ \downarrow +\frac{1}{4} \\ (x-3)^2-9+5 \\ (x-3)^2-4 \end{aligned}$$

Let $x-3 = 2 \sec \theta \rightarrow \frac{x-3}{2} = \sec \theta$
 $dx = 2 \sec \theta \tan \theta d\theta$
 $x = 2 \sec \theta + 3$



$$\int \frac{2(2 \sec \theta + 3) - 1}{\sqrt{(2 \sec \theta)^2 - 4}} \cdot 2 \sec \theta \tan \theta d\theta$$

$$\rightarrow \sqrt{4 \sec^2 \theta - 4} = \sqrt{4(\sec^2 \theta - 1)} = \sqrt{4 \tan^2 \theta} = 2 \tan \theta$$

$$\int \frac{4 \sec \theta + 6 - 1}{2 \tan \theta} \cdot 2 \sec \theta \tan \theta d\theta$$

$$\int (4 \sec \theta + 5) \sec \theta d\theta$$

$$\int (4 \sec^2 \theta + 5 \sec \theta) d\theta$$

$$= 4 \tan \theta + 5 \ln |\sec \theta + \tan \theta| + C$$

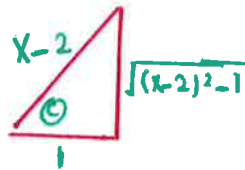
$$= 4 \frac{\sqrt{(x-3)^2-4}}{2} + 5 \ln \left| \frac{x-3}{2} + \frac{\sqrt{(x-3)^2-4}}{2} \right| + C$$

$$\int \frac{x}{(x-2)\sqrt{x^2-4x+3}} dx$$

$$\int \frac{x}{(x-2)\sqrt{(x-2)^2-1}} dx$$

$$\begin{aligned} x^2 - 4x + 3 \\ \downarrow +\frac{1}{2} \\ (x-2)^2 - 4 + 3 \\ \text{مربع} \\ (x-2)^2 - 1 \end{aligned}$$

Let $\begin{cases} x-2 = \sec\theta \\ dx = \sec\theta \tan\theta d\theta \end{cases}$



$$x = \sec\theta + 2$$

$$\int \frac{\sec\theta + 2}{\sec\theta \sqrt{\sec^2\theta - 1}} \sec\theta \tan\theta d\theta$$

$$\sqrt{\tan^2\theta} = \tan\theta$$

$$\int \frac{\sec\theta + 2}{\sec\theta} \sec\theta \tan\theta d\theta$$

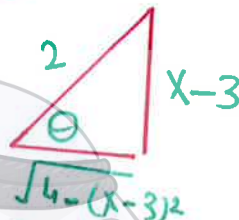
$$\begin{aligned} & \int (\sec\theta + 2) d\theta \\ &= \ln|\sec\theta + \tan\theta| + 2\theta + C \end{aligned}$$

$$= \ln \left| \frac{x-2}{1} + \frac{\sqrt{(x-2)^2-1}}{1} \right| + 2 \sec^{-1}(x-2) + C$$

$$\int \sqrt{6x - 5 - x^2} dx$$

$$\int \sqrt{4 - (x-3)^2} dx$$

Let $x-3 = 2 \sin \theta \rightarrow \frac{x-3}{2} = \sin \theta$
 $dx = 2 \cos \theta d\theta$



$$\begin{aligned} & - [x^2 - 6x + 5] \\ & \quad \downarrow + \frac{1}{2} \\ & - [(x-3)^2 - 9 + 5] \\ & - [(x-3)^2 - 4] \\ & \quad 4 - (x-3)^2 \end{aligned}$$

$$\int \sqrt{4 - (2 \sin \theta)^2} \cdot 2 \cos \theta d\theta$$

$$\rightarrow \sqrt{4 - 4 \sin^2 \theta} = \sqrt{4(1 - \sin^2 \theta)} = \sqrt{4 \cos^2 \theta} = 2 \cos \theta$$

$$\int 2 \cos \theta \cdot 2 \cos \theta d\theta$$

$$4 \int \cos^2 \theta d\theta$$

$$4 \cdot \frac{1}{2} \int (1 + \cos 2\theta) d\theta$$

$$2 \left[\theta + \frac{\sin 2\theta}{2} \right] + C$$

$$2 \left[\theta + \frac{2 \sin \theta \cos \theta}{2} \right] + C$$

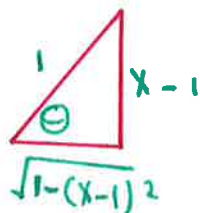
$$2 \left[\sin^{-1} \left(\frac{x-3}{2} \right) + \frac{x-3}{2} \cdot \frac{\sqrt{4 - (x-3)^2}}{2} \right] + C$$

$$\int x \sqrt{2x - x^2} dx$$

$$\int x \sqrt{1 - (x-1)^2} dx$$

Let $\begin{cases} x-1 = \sin \theta \\ dx = \cos \theta d\theta \end{cases}$

$$x = \sin \theta + 1$$



$$\begin{aligned} & - [x^2 - 2x] \\ & - [(x-1)^2 - 1] \\ & 1 - (x-1)^2 \end{aligned}$$

$$\int (\sin \theta + 1) \sqrt{1 - \sin^2 \theta} \cdot \cos \theta d\theta$$

$$\int (\sin \theta + 1) \cos \theta \cos \theta d\theta = \cos \theta$$

$$\int (\sin \theta \cos^2 \theta + \cos^2 \theta) d\theta$$

$$\int \sin \theta \cos^2 \theta d\theta + \int \cos^2 \theta d\theta$$

Let $\begin{cases} u = \cos \theta \\ du = -\sin \theta d\theta \\ -du = \sin \theta d\theta \end{cases}$

$$\begin{aligned} \int u^2 du &= -\frac{u^3}{3} + C_1 \\ &= -\frac{\cos^3 \theta}{3} + C_1 \end{aligned}$$

$$\begin{aligned} & \frac{1}{2} \int (1 + \cos 2\theta) d\theta \\ & \frac{1}{2} \left[\theta + \frac{\sin 2\theta}{2} \right] + C_2 \\ & \frac{2 \sin \theta \cos \theta}{2} \end{aligned}$$

$$\Rightarrow -\frac{\left(\frac{\sqrt{1-(x-1)^2}}{1}\right)^3}{3} + \frac{1}{2} \left[\sin^{-1}(x-1) + \frac{x-1}{1} \cdot \frac{\sqrt{1-(x-1)^2}}{1} \right]$$

$$\int \frac{x dx}{\sqrt{x^2 - 4x + 3}}$$

Nov 2016

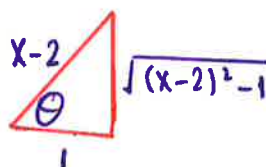
$$\int \frac{x}{\sqrt{(x-2)^2 - 1}} dx$$

$$\begin{aligned} x^2 - 4x + 3 \\ \downarrow + \frac{1}{4} \\ (x-2)^2 - 4 + 3 \\ \downarrow \\ (x-2)^2 - 1 \end{aligned}$$

Let

$$x-2 = \sec \theta$$

$$dx = \sec \theta \tan \theta d\theta$$



$$x = \sec \theta + 2$$

$$\int \frac{\sec \theta + 2}{\sqrt{\sec^2 \theta - 1}} \cdot \sec \theta \tan \theta d\theta$$

$$\int \frac{\sec \theta + 2}{\tan \theta} \sec \theta \tan \theta d\theta$$

$\sqrt{\tan^2 \theta} = \tan \theta$

$$\int (\sec^2 \theta + 2 \sec \theta) d\theta$$

$$= \tan \theta + 2 \ln |\sec \theta + \tan \theta| + C$$

$$= \frac{\sqrt{(x-2)^2 - 1}}{1} + 2 \ln \left| \frac{x-2}{1} + \frac{\sqrt{(x-2)^2 - 1}}{1} \right| + C$$

$$\int \frac{x+5}{\sqrt{x^2-6x+10}} dx$$

Fall 2022

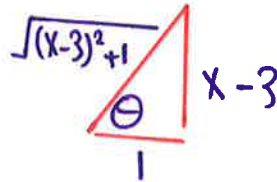
$$\int \frac{x+5}{\sqrt{(x-3)^2+1}} dx$$

$$\begin{aligned} x^2-6x+10 \\ \downarrow +\frac{1}{4} \\ (x-3)^2-9+10 \\ \downarrow \\ (x-3)^2+1 \end{aligned}$$

Let

$$x-3 = \tan \theta$$

$$dx = \sec^2 \theta d\theta$$



$$x = \tan \theta + 3$$

$$\int \frac{\tan \theta + 3 + 5}{\sqrt{\tan^2 \theta + 1}} \cdot \sec^2 \theta d\theta$$

$$\int \frac{\tan \theta + 8}{\cancel{\sec \theta}} \cdot \sec \theta d\theta$$

$$\int (\sec \theta \tan \theta + 8 \sec \theta) d\theta$$

$$= \sec \theta + 8 \ln |\sec \theta + \tan \theta| + C$$

$$= \frac{\sqrt{(x-3)^2+1}}{1} + 8 \ln \left| \frac{\sqrt{(x-3)^2+1}}{1} + \frac{x-3}{1} \right| + C$$

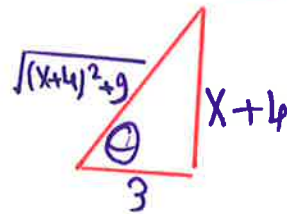
$$\int \frac{(x+4)^3}{\sqrt{x^2+8x+25}} dx$$

Oct 2017

$$\int \frac{(x+4)^3}{\sqrt{(x+4)^2+9}} dx$$

$$\begin{aligned} x^2+8x+25 \\ \downarrow +16 \\ (x+4)^2-16+25 \\ \downarrow \\ (x+4)^2+9 \end{aligned}$$

Let $x+4 = 3 \tan \theta \rightarrow \frac{x+4}{3} = \tan \theta$
 $dx = 3 \sec^2 \theta d\theta$



$$\int \frac{(3 \tan \theta)^3}{\sqrt{(3 \tan \theta)^2+9}} \cdot 3 \sec^2 \theta d\theta$$

$$\int \frac{27 \tan^3 \theta}{\cancel{3 \sec \theta}} \cdot \cancel{3 \sec^2 \theta} d\theta$$

$\sqrt{9 \tan^2 \theta + 9} = \sqrt{9(\tan^2 \theta + 1)} = \sqrt{9 \sec^2 \theta} = 3 \sec \theta$

$$27 \int \tan^3 \theta \sec \theta d\theta$$

$$27 \int \tan^2 \theta \sec \theta \tan \theta d\theta$$

$$27 \int (\sec^2 \theta - 1) \sec \theta \tan \theta d\theta$$

Let $u = \sec \theta$

$$du = \sec \theta \tan \theta d\theta$$

$$27 \int (u^2 - 1) du = 27 \left[\frac{u^3}{3} - u \right] + c$$

$$= 27 \left[\frac{(\sec \theta)^3}{3} - \sec \theta \right] + c$$

$$= 27 \left[\frac{\left(\frac{\sqrt{(x+4)^2+9}}{3} \right)^3}{3} - \frac{\sqrt{(x+4)^2+9}}{3} \right] + c$$

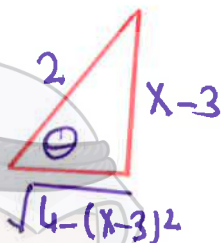
$$\int \frac{(x-3)^2}{\sqrt{6x-5-x^2}} dx$$

$$\int \frac{(x-3)^2}{\sqrt{4-(x-3)^2}} dx$$

$$\begin{aligned} & - [x^2 - 6x + 5] \\ & - [(x-3)^2 - 9 + 5] \\ & - [(x-3)^2 - 4] \\ & \quad 4 - (x-3)^2 \end{aligned}$$

Let $x-3 = 2 \sin \theta \rightarrow \frac{x-3}{2} = \sin \theta$
 $dx = 2 \cos \theta d\theta$

$$\int \frac{(2 \sin \theta)^2}{\sqrt{4-(2 \sin \theta)^2}} \cdot 2 \cos \theta d\theta$$



$$\int \frac{4 \sin^2 \theta}{\sqrt{4-4 \sin^2 \theta}} \cdot 2 \cos \theta d\theta$$

$$\int \frac{4 \sin^2 \theta}{2 \cos \theta} \cdot 2 \cos \theta d\theta$$

$\sqrt{4(1-\sin^2 \theta)} = \sqrt{4 \cos^2 \theta} = 2 \cos \theta$

$$4 \int \sin^2 \theta d\theta$$

$$4 \cdot \frac{1}{2} \int (1 - \cos 2\theta) d\theta$$

$$2 \left[\theta - \frac{\sin 2\theta}{2} \right] + C$$

$$2 \left[\theta - \frac{2 \sin \theta \cos \theta}{2} \right] + C$$

$$2 \left[\sin^{-1} \left(\frac{x-3}{2} \right) - \frac{x-3}{2} \cdot \frac{\sqrt{4-(x-3)^2}}{2} \right] + C$$

$$\int \frac{x^2 + 6x + 9}{\sqrt{7 - 6x - x^2}} dx$$

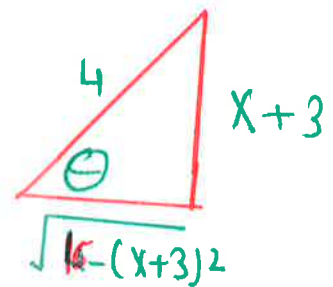
$$\int \frac{(x+3)(x+3)}{\sqrt{16 - (x+3)^2}} dx$$

$$\int \frac{(x+3)^2}{\sqrt{16 - (x+3)^2}} dx$$

$$\begin{aligned} & - [x^2 + 6x - 7] \\ & - [(x+3)^2 - 9 - 7] \\ & - [(x+3)^2 - 16] \\ & 16 - (x+3)^2 \end{aligned}$$

Let $x+3 = 4 \sin \theta \rightarrow \frac{x+3}{4} = \sin \theta$
 $dx = 4 \cos \theta d\theta$

$$\int \frac{(4 \sin \theta)^2}{\sqrt{16 - (4 \sin \theta)^2}} \cdot 4 \cos \theta d\theta$$



$$\begin{aligned} \sqrt{16 - 16 \sin^2 \theta} &= \sqrt{16(1 - \sin^2 \theta)} = \sqrt{16 \cos^2 \theta} \\ &= 4 \cos \theta \end{aligned}$$

$$\int \frac{16 \sin^2 \theta}{4 \cos \theta} \cdot 4 \cos \theta d\theta$$

$$16 \int \sin^2 \theta d\theta$$

$$16 \cdot \frac{1}{2} \int (1 - \cos 2\theta) d\theta$$

$$8 \left[\theta - \frac{\sin 2\theta}{2} \right] + C$$

$$8 \left[\theta - \frac{2 \sin \theta \cos \theta}{2} \right] + C$$

$$8 \left[\sin^{-1} \left(\frac{x+3}{4} \right) - \frac{x+3}{4} \cdot \frac{\sqrt{16 - (x+3)^2}}{4} \right] + C$$

$$\int (1+x)e^x \sqrt{1-(xe^x)^2} dx$$

Let $u = xe^x$

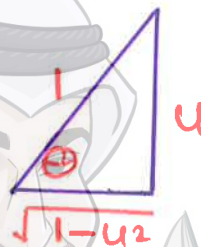
$$du = (1)e^x + xe^x dx$$

$$du = e^x (1+x) dx$$

$$\int \sqrt{1-u^2} du$$

Let $u = \sin \theta$

$$du = \cos \theta d\theta$$



$$\int \sqrt{1-\sin^2 \theta} \cos \theta d\theta$$

$\swarrow$
 $\sqrt{\cos^2 \theta} = \cos \theta$

$$\int \cos \theta \cos \theta d\theta = \int \cos^2 \theta d\theta$$

$$\frac{1}{2} \int (1 + \cos 2\theta) d\theta$$

$$\frac{1}{2} \left[\theta + \frac{\sin 2\theta}{2} \right] + C$$

$$\frac{1}{2} \left[\theta + \frac{2 \sin \theta \cos \theta}{2} \right] + C$$

$$\frac{1}{2} \left[\sin^{-1} u + \frac{u}{1} \cdot \frac{1}{1} \sqrt{1-u^2} \right] + C = \frac{1}{2} \left[\sin^{-1}(xe^x) + xe^x \sqrt{1-(xe^x)^2} \right] + C$$

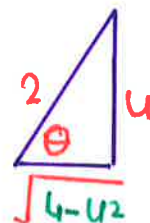
$$\int \frac{\ln^2 x}{x\sqrt{4 - \ln^2 x}} dx$$

$$\int \frac{\ln^2 x}{x\sqrt{4 - (\ln x)^2}} dx$$

Let $u = \ln x$
 $du = \frac{1}{x} dx$

$$\int \frac{u^2}{\sqrt{4 - u^2}} du$$

Let $u = 2 \sin \theta \rightarrow \frac{u}{2} = \sin \theta$
 $du = 2 \cos \theta d\theta$



$$\int \frac{(2 \sin \theta)^2}{\sqrt{4 - (2 \sin \theta)^2}} \cdot 2 \cos \theta d\theta$$

$$\begin{aligned} \sqrt{4 - 4 \sin^2 \theta} &= \sqrt{4(1 - \sin^2 \theta)} = \sqrt{4 \cos^2 \theta} \\ &= 2 \cos \theta \end{aligned}$$

$$\int \frac{4 \sin^2 \theta}{\cancel{2 \cos \theta}} \cdot \cancel{2 \cos \theta} d\theta$$

$$4 \cdot \frac{1}{2} \int (1 - \cos 2\theta) d\theta$$

$$2 \left[\theta - \frac{\sin 2\theta}{2} \right] + C$$

$$2 \left[\theta - \frac{2 \sin \theta \cos \theta}{2} \right] + C$$

$$2 \left[\sin^{-1} \left(\frac{u}{2} \right) - \frac{u}{2} \cdot \frac{\sqrt{4 - u^2}}{2} \right] + C$$

$$2 \left[\sin^{-1} \left(\frac{\ln x}{2} \right) - \frac{\ln x}{2} \cdot \frac{\sqrt{4 - (\ln x)^2}}{2} \right] + C$$

$$\int \frac{\cosh x}{\sqrt{\sinh^2 x - 3}} dx$$

$$\int \frac{\cosh x}{\sqrt{(\sinh x)^2 - 3}} dx$$

Let $u = \sinh x$
 $du = \cosh x dx$

$$\int \frac{1}{\sqrt{u^2 - 3}} du$$

Let $u = \sqrt{3} \sec \theta \rightarrow \frac{u}{\sqrt{3}} = \sec \theta$
 $du = \sqrt{3} \sec \theta \tan \theta d\theta$

$$\int \frac{1}{\sqrt{(\sqrt{3} \sec \theta)^2 - 3}} \cdot \sqrt{3} \sec \theta \tan \theta d\theta$$

$\sqrt{(\sqrt{3} \sec \theta)^2 - 3} = \sqrt{3 \sec^2 \theta - 3} = \sqrt{3(\sec^2 \theta - 1)} = \sqrt{3 \tan^2 \theta} = \sqrt{3} \tan \theta$

$$\int \frac{1}{\sqrt{3} \tan \theta} \cdot \sqrt{3} \sec \theta \tan \theta d\theta = \int \sec \theta d\theta$$

$$\begin{aligned} \int \sec \theta d\theta &= \ln |\sec \theta + \tan \theta| + C \\ &= \ln \left| \frac{u}{\sqrt{3}} + \frac{\sqrt{u^2 - 3}}{\sqrt{3}} \right| + C \\ &= \ln \left| \frac{\sinh x}{\sqrt{3}} + \frac{\sqrt{(\sinh x)^2 - 3}}{\sqrt{3}} \right| + C \end{aligned}$$

$$\int \frac{\cos x}{(\sin^2 x + 8 \sin x + 7)^{\frac{3}{2}}} dx$$

$$\int \frac{\cos x}{[(\sin x)^2 + 8 \sin x + 7]^{\frac{3}{2}}} dx$$

Let $u = \sin x$
 $du = \cos x \, dx$

$$\int \frac{1}{[u^2 + 8u + 7]^{\frac{3}{2}}} du$$

$$\int \frac{1}{[(u+4)^2 - 9]^{\frac{3}{2}}} du$$

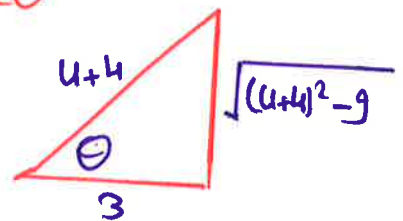
$$u^2 + 8u + 7$$

$$(u+4)^2 - 16 + 7$$

$$(u+4)^2 - 9$$

Let $u+4 = 3 \sec \theta \rightarrow \frac{u+4}{3} = \sec \theta$
 $du = 3 \sec \theta \tan \theta \, d\theta$

$$\int \frac{1}{[(3 \sec \theta)^2 - 9]^{\frac{3}{2}}} \cdot 3 \sec \theta \tan \theta \, d\theta$$



$$\begin{aligned} & \rightarrow (9 \sec^2 \theta - 9)^{\frac{3}{2}} = (9(\sec^2 \theta - 1))^{\frac{3}{2}} \\ & = (9 \tan^2 \theta)^{\frac{3}{2}} = (3 \tan \theta)^3 \\ & = 3^3 \tan^3 \theta \end{aligned}$$

$$\int \frac{1}{3^3 \tan^3 \theta} \cdot 3 \sec \theta \tan \theta \, d\theta$$

$$\int \frac{\sec \theta}{3^2 \tan^2 \theta} d\theta$$

$$\frac{1}{9} \int \frac{\cancel{\cos \theta}^1}{\sin^2 \theta} d\theta$$

$$\frac{1}{9} \int \frac{\cos \theta}{\sin^2 \theta} d\theta$$

$$\frac{1}{9} \int \underline{\cos \theta} (\sin \theta)^{-2} d\theta$$

Let $z = \sin \theta$
 $dz = \cos \theta d\theta$

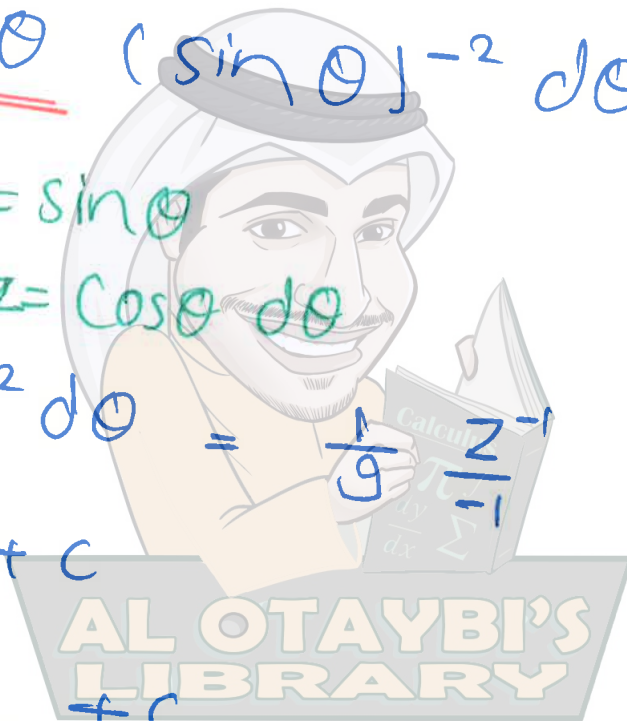
$$\frac{1}{9} \int z^{-2} d\theta = \frac{1}{9} \frac{z^{-1}}{-1} + C$$

$$\frac{1}{9} \frac{1}{z} + C$$

$$\frac{1}{9} \frac{1}{\sin \theta} + C$$

$$\frac{1}{9} \frac{u+4}{\sqrt{(u+4)^2-9}} + C$$

$$\frac{1}{9} \frac{\sin x + 4}{\sqrt{(\sin x + 4)^2 - 9}} + C$$



$$\int \frac{\sqrt{e^{2x} - 2e^x - 8}}{1 - e^{-x}} dx$$

$$\int \frac{\sqrt{(e^x)^2 - 2e^x - 8}}{1 - e^{-x}} dx$$

Let $u = e^x$
 $du = e^x dx$

$$\int \frac{\sqrt{(e^x)^2 - 2e^x - 8}}{1 - e^{-x}} dx$$

$\frac{e^x}{e^x}$

$$\int \frac{\sqrt{(e^x)^2 - 2e^x - 8}}{e^x - 1} e^x dx$$

$$\int \frac{\sqrt{u^2 - 2u - 8}}{u - 1} du$$

$$u^2 - 2u - 8$$

$$(u - 1)^2 - 1 - 8$$

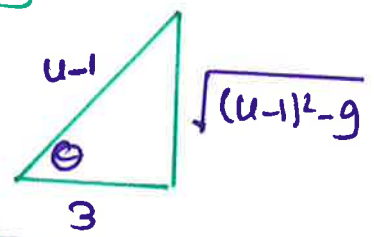
$$(u - 1)^2 - 9$$

$$\int \frac{\sqrt{(u-1)^2 - 9}}{u-1} du$$

Let $u - 1 = 3 \sec \theta \rightarrow \frac{u-1}{3} = \sec \theta$
 $\frac{du}{d\theta} = 3 \sec \theta \tan \theta$

$$\int \frac{\sqrt{(3 \sec \theta)^2 - 9}}{3 \sec \theta} \cdot 3 \sec \theta \tan \theta d\theta$$

$$\sqrt{9 \sec^2 \theta - 9} = \sqrt{9(\sec^2 \theta - 1)} = \sqrt{9 \tan^2 \theta} = 3 \tan \theta$$



$$\int \frac{3 \tan \theta}{1} \cdot \tan \theta \, d\theta$$

$$3 \int \tan^2 \theta \, d\theta$$

$$3 \int (\sec^2 \theta - 1) \, d\theta$$

$$3 [\tan \theta - \theta] + C$$

$$3 \left[\frac{\sqrt{(u-1)^2 - 9}}{3} - \sec^{-1} \left(\frac{u-1}{3} \right) \right]$$

$$3 \left[\frac{\sqrt{(e^x - 1)^2 - 9}}{3} - \sec^{-1} \left(\frac{e^x - 1}{3} \right) \right] + C$$

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Text Book Question



$$\int_0^{\frac{\pi}{2}} \frac{\cos t}{\sqrt{1 + \sin^2 t}} dt$$

Let $u = \sin t$
 $du = \cos t dt$

$$\int \frac{1}{\sqrt{1 + u^2}} du$$

Let $u = \tan \theta$
 $du = \sec^2 \theta d\theta$

$$\int \frac{1}{\sqrt{1 + \tan^2 \theta}} \cdot \sec^2 \theta d\theta$$

$\sqrt{1 + \tan^2 \theta} = \sec \theta$
 $\int \frac{1}{\sec \theta} \cdot \sec^2 \theta d\theta$

$$= \ln |\sec \theta + \tan \theta|$$

$$= \ln \left| \frac{\sqrt{u^2 + 1}}{1} + \frac{u}{1} \right|$$

$$= \left[\ln \left| \sqrt{\sin^2 t + 1} + \sin t \right| \right]_0^{\frac{\pi}{2}}$$

$$= \ln |\sqrt{1+1} + 1| - \ln |\sqrt{0+1} + 0|$$

$$= \ln |\sqrt{2} + 1| - 0$$



Text Book Question



$$\int \frac{x^2}{\sqrt{9 - 25x^2}} dx$$

$$\int \frac{x^2}{\sqrt{25(\frac{9}{25} - x^2)}} dx$$



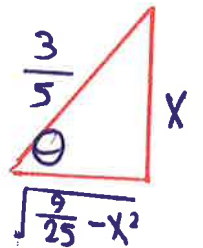
Tricky Question !!



$$\frac{1}{\sqrt{25}} \int \frac{x^2}{\sqrt{\frac{9}{25} - x^2}} dx$$

Let $x = \frac{3}{5} \sin \theta \rightarrow \frac{x}{\frac{3}{5}} = \sin \theta$

$$dx = \frac{3}{5} \cos \theta d\theta$$



$$\frac{1}{5} \int \frac{(\frac{3}{5} \sin \theta)^2}{\sqrt{\frac{9}{25} - \frac{9}{25} \sin^2 \theta}} \cdot \frac{3}{5} \cos \theta d\theta$$

$$\frac{1}{5} \int \frac{\frac{9}{25} \sin^2 \theta}{\sqrt{\frac{9}{25} (1 - \sin^2 \theta)}} \cdot \frac{3}{5} \cos \theta d\theta = \frac{1}{5} \int \frac{\frac{9}{25} \sin^2 \theta}{\frac{3}{5} \cos \theta} \cdot \frac{3}{5} \cos \theta d\theta$$

$$\frac{1}{5} \cdot \frac{9}{25} \cdot \frac{1}{2} \int (1 - \cos 2\theta) d\theta$$

$$\frac{9}{250} \left[\theta - \frac{\sin 2\theta}{2} \right] + C$$

$$\frac{9}{250} \left[\theta - \frac{2 \sin \theta \cos \theta}{2} \right] + C$$

$$\frac{9}{250} \left[\sin^{-1}\left(\frac{x}{\frac{3}{5}}\right) - \frac{x}{\frac{3}{5}} \cdot \frac{\sqrt{\frac{9}{25} - x^2}}{\frac{3}{5}} \right] + C$$

$$\int \frac{x^2}{(4x^2 + 9)^{\frac{3}{2}}} dx$$

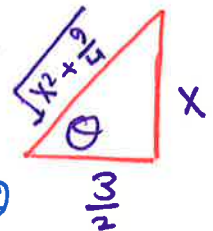
$$\int \frac{x^2}{[4(x^2 + \frac{9}{4})]^{\frac{3}{2}}} dx$$

$$\int \frac{x^2}{4^{\frac{3}{2}} (x^2 + \frac{9}{4})^{\frac{3}{2}}} dx$$

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x²
لأنه

Let $x = \frac{3}{2} \tan \theta \rightarrow \frac{x}{\frac{3}{2}} = \tan \theta$

$$dx = \frac{3}{2} \sec^2 \theta d\theta$$



$$\frac{1}{4^{\frac{3}{2}}} \int \frac{(\frac{3}{2} \tan \theta)^2 \cdot \frac{3}{2} \sec^2 \theta d\theta}{\left((\frac{3}{2} \tan \theta)^2 + \frac{9}{4} \right)^{\frac{3}{2}}}$$

$$\begin{aligned} &\rightarrow \left(\frac{9}{4} \tan^2 \theta + \frac{9}{4} \right)^{\frac{3}{2}} = \left(\frac{9}{4} (\tan^2 \theta + 1) \right)^{\frac{3}{2}} \\ &= \left(\frac{9}{4} \sec^2 \theta \right)^{\frac{3}{2}} = \left(\frac{9}{4} \right)^{\frac{3}{2}} (\sec^2 \theta)^{\frac{3}{2}} \\ &= \left(\frac{3}{2} \right)^3 \sec^3 \theta \end{aligned}$$

$$\frac{1}{4^{\frac{3}{2}}} \int \frac{\frac{9}{4} \tan^2 \theta}{\left(\frac{3}{2} \right)^3 \sec^3 \theta} \cdot \frac{3}{2} \sec^2 \theta d\theta$$

$$\frac{1}{4^{\frac{3}{2}}} \int \frac{\frac{9}{4} \tan^2 \theta}{\left(\frac{3}{2} \right)^2 \sec \theta} d\theta$$

$$\frac{1}{4^{\frac{3}{2}}} \int \frac{\tan^2 \theta}{\sec \theta} d\theta$$

$\swarrow$
 2^3

$$\frac{1}{8} \int \frac{\frac{\sin^2 \theta}{\cancel{\cos^2 \theta}}}{\cancel{\cos \theta}} d\theta$$

$$\frac{1}{8} \int \frac{\sin^2 \theta}{\cos \theta} d\theta$$

$$\frac{1}{8} \int \frac{1 - \cos^2 \theta}{\cos \theta} d\theta$$

$$\frac{1}{8} \int \left[\frac{1}{\cos \theta} - \frac{\cos^2 \theta}{\cos \theta} \right] d\theta$$

$$\frac{1}{8} \int [\sec \theta - \cos \theta] d\theta$$

$$\frac{1}{8} [\ln |\sec \theta + \tan \theta| - \sin \theta] + C$$

$$\frac{1}{8} \left[\ln \left| \frac{\sqrt{x^2 + \frac{2}{4}}}{\frac{x}{2}} + \frac{x}{\frac{3}{2}} \right| - \frac{x}{\sqrt{x^2 + \frac{2}{4}}} \right] + C$$



Text Book Question



Tricky Question !!



$$\int \frac{x^2}{(3 + 4x - 4x^2)^{\frac{3}{2}}} dx$$

$$\int \frac{x^2}{(4[1 - (x - \frac{1}{2})^2])^{\frac{3}{2}}} dx$$

$$\begin{aligned} &= 4^{\frac{3}{2}} [1 - (x - \frac{1}{2})^2]^{\frac{3}{2}} \\ &= 2^3 [1 - (x - \frac{1}{2})^2]^{\frac{3}{2}} \end{aligned}$$

$$\int \frac{x^2}{8 [1 - (x - \frac{1}{2})^2]^{\frac{3}{2}}} dx$$

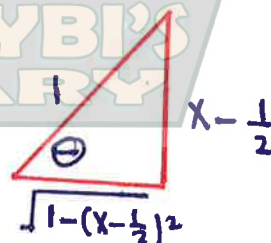
$$\frac{1}{8} \int \frac{x^2}{[1 - (x - \frac{1}{2})^2]^{\frac{3}{2}}} dx$$

$$\begin{aligned} &= -4 [x^2 - x - \frac{3}{4}] \\ &= -4 [(x - \frac{1}{2})^2 - \frac{1}{4} - \frac{3}{4}] \\ &= -4 [(x - \frac{1}{2})^2 - 1] \\ &= 4 [1 - (x - \frac{1}{2})^2] \end{aligned}$$

Let $x - \frac{1}{2} = \sin \theta$

$dx = \cos \theta d\theta$

$x = \sin \theta + \frac{1}{2}$



$$\frac{1}{8} \int \frac{(\sin \theta + \frac{1}{2})^2}{[1 - \sin^2 \theta]^{\frac{3}{2}}} \cdot \cos \theta d\theta$$

$$\frac{1}{8} \int \frac{\sin^2 \theta + \sin \theta + \frac{1}{4}}{\cos^3 \theta} \cos \theta d\theta$$

$(\cos^2 \theta)^{\frac{3}{2}} = \cos^3 \theta$

$$\frac{1}{8} \int \frac{\sin^2 \theta + \sin \theta + \frac{1}{4}}{\cos^2 \theta} d\theta$$

$$\frac{1}{8} \int \left[\frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\sin \theta}{\cos^2 \theta} + \frac{\frac{1}{4}}{\cos^2 \theta} \right] d\theta$$

$$\int \tan^2 \theta d\theta$$

$$\int (\sec^2 \theta - 1) d\theta$$

$$\tan \theta - \theta + C_1$$

$$\int \underline{\sin \theta} (\cos \theta)^{-2} d\theta$$

$$u = \cos \theta$$

$$du = -\sin \theta d\theta$$

$$-du = \sin \theta d\theta$$

$$-\int u^{-2} du$$

$$-\frac{u^{-1}}{-1} + C_2$$

$$\frac{1}{u} + C_2$$

$$\frac{1}{\cos \theta} + C_2$$

$$\frac{1}{4} \int \sec^2 \theta d\theta$$

$$\frac{1}{4} \tan \theta + C_3$$

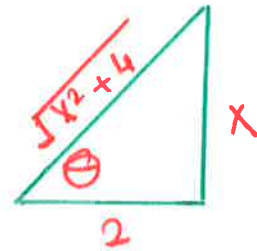
$$\frac{1}{8} \left[\tan \theta - \theta + \frac{1}{\cos \theta} + \frac{1}{4} \tan \theta \right] + C$$

$$\frac{1}{8} \left[\frac{x - \frac{1}{2}}{\sqrt{1 - (x - \frac{1}{2})^2}} - \sin^{-1}(x - \frac{1}{2}) + \frac{1}{\sqrt{1 - (x - \frac{1}{2})^2}} + \frac{1}{4} \frac{x - \frac{1}{2}}{\sqrt{1 - (x - \frac{1}{2})^2}} \right] + C$$



$$\int \frac{1}{(x^2 + 4)^2} dx$$

Let $x = 2 \tan \theta \longrightarrow \frac{x}{2} = \tan \theta$
 $dx = 2 \sec^2 \theta d\theta$



$$\int \frac{1}{(2 \tan \theta)^2 + 4)^2} \cdot 2 \sec^2 \theta d\theta$$

$$\longrightarrow (4 \tan^2 \theta + 4)^2 = (4(\tan^2 \theta + 1)) = 4 \sec^2 \theta$$

$$\int \frac{1}{(4 \sec^2 \theta)^2} \cdot 2 \sec^2 \theta d\theta$$

$$\int \frac{1}{16 \sec^4 \theta} \cdot 2 \sec^2 \theta d\theta$$

$$\int \frac{1}{8 \sec^2 \theta} d\theta = \frac{1}{8} \int \cos^2 \theta d\theta$$

$$\frac{1}{8} \cdot \frac{1}{2} \int (1 + \cos 2\theta) d\theta$$

$$\frac{1}{16} \left[\theta + \frac{\sin 2\theta}{2} \right] + C$$

$$\frac{1}{16} \left[\theta + \frac{2 \sin \theta \cos \theta}{2} \right] + C$$

$$\frac{1}{16} \left[\tan^{-1}\left(\frac{x}{2}\right) + \frac{x}{\sqrt{x^2 + 4}} \cdot \frac{2}{\sqrt{x^2 + 4}} \right] + C$$



Text Book Question

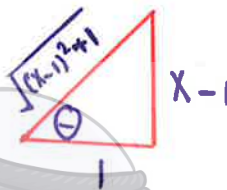


$$\int \frac{x^2 + 1}{(x^2 - 2x + 2)^2} dx$$

$$\int \frac{x^2 + 1}{[(x-1)^2 + 1]^2} dx$$

$$\begin{aligned} x^2 - 2x + 2 \\ (x-1)^2 - 1 + 2 \\ (x-1)^2 + 1 \end{aligned}$$

Let $x-1 = \tan \theta$
 $dx = \sec^2 \theta d\theta$



$$x = \tan \theta + 1$$

$$\int \frac{(\tan \theta + 1)^2 + 1}{[\tan^2 \theta + 1]^2} \cdot \sec^2 \theta d\theta$$

$$(\sec^2 \theta)^2 = \sec^4 \theta$$

$$\int \frac{\tan^2 \theta + 2 \tan \theta + 1 + 1}{\sec^4 \theta} \sec^2 \theta d\theta$$

$$\int \frac{\tan^2 \theta + 2 \tan \theta + 2}{\sec^2 \theta} d\theta$$

$$\int \left[\frac{\tan^2 \theta}{\sec^2 \theta} + \frac{2 \tan \theta}{\sec^2 \theta} + \frac{2}{\sec^2 \theta} \right] d\theta$$

$$\int \frac{\frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{1}{\cos^2 \theta}} d\theta$$

$$+ \int \frac{2 \frac{\sin \theta}{\cos \theta}}{\frac{1}{\cos^2 \theta}} d\theta$$

$$+ \int 2 \cos^2 \theta d\theta$$

$$\frac{1}{2} \int (1 - \cos 2\theta) d\theta$$

$$\frac{1}{2} \left[\theta - \frac{\sin 2\theta}{2} \right] + C_1$$

$$\frac{1}{2} \left[\theta - \frac{2 \sin \theta \cos \theta}{2} \right] + C_1$$

$$2 \int \sin \theta \cos \theta d\theta$$

$$u = \sin \theta \rightarrow du = \cos \theta d\theta$$

$$2 \int u du = 2 \frac{u^2}{2} + C_2 = \sin^2 \theta + C_2$$

$$\frac{1}{2} \int 2(1 + \cos 2\theta) d\theta$$

$$\theta + \frac{\sin 2\theta}{2} + C_3$$

$$\theta + \frac{2 \sin \theta \cos \theta}{2} + C_3$$

ثم نكتب الجواب نهائياً بـ صيغة x عن طريق التمثيل



أخوووو الدرس



$$\int \frac{x}{(x^4 + 4)^2} dx$$

$$\int \frac{x}{((x^2)^2 + 4)^2} dx$$

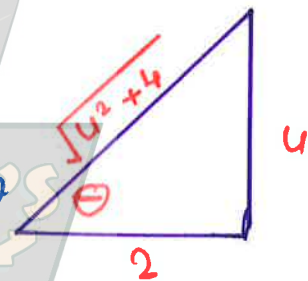
Let $u = x^2$

$$du = 2x dx \rightarrow \frac{1}{2} du = x dx$$

$$\frac{1}{2} \int \frac{1}{(u^2 + 4)^2} du$$

Let $u = 2 \tan \theta \rightarrow \frac{u}{2} = \tan \theta$
 $du = 2 \sec^2 \theta d\theta$

$$\frac{1}{2} \int \frac{1}{((2 \tan \theta)^2 + 4)^2} 2 \sec^2 \theta d\theta$$



$$\rightarrow 4 \tan^2 \theta + 4 = 4(\tan^2 \theta + 1) = 4 \sec^2 \theta$$

$$\frac{1}{2} \int \frac{1}{(4 \sec^2 \theta)^2} \cdot 2 \sec^2 \theta d\theta$$

$$\frac{1}{2} \int \frac{1}{16 \sec^4 \theta} 2 \sec^2 \theta d\theta$$

$$\frac{1}{16} \int \cos^2 \theta d\theta = \frac{1}{16} \cdot \frac{1}{2} \int (1 + \cos 2\theta) d\theta$$

$$\frac{1}{32} \left[\theta + \frac{\sin 2\theta}{2} \right] + C = \frac{1}{32} \left[\theta + \frac{2 \sin \theta \cos \theta}{2} \right]$$

$$= \frac{1}{32} \left[\tan^{-1}\left(\frac{u}{2}\right) + \frac{u}{\sqrt{u^2 + 4}} \cdot \frac{2}{\sqrt{u^2 + 4}} \right] + C = \frac{1}{32} \left[\tan^{-1}\left(\frac{x^2}{2}\right) + \frac{2x^2}{(x^2)^2 + 4} \right] + C$$



Calculus B Quiz # 4



$$\int \cos(5x) dx = \frac{\sin(5x)}{5} + C$$

$$\int e^{3x} dx = \frac{e^{3x}}{3} + C$$

$$\int \left(\frac{1}{\sqrt{2-x^2}} + \frac{1}{x\sqrt{x^2-2}} + \frac{x}{\sqrt{x^2-2}} \right) dx$$

$$\sin^{-1}\left(\frac{x}{\sqrt{2}}\right) + C_1$$

$$\frac{1}{\sqrt{2}} \sec^{-1}\left(\frac{x}{\sqrt{2}}\right) + C_2$$

$$\int x (x^2-2)^{-\frac{1}{2}} dx$$

$$\text{Let } u = x^2 - 2$$

$$du = 2x dx$$

$$\frac{1}{2} du = x dx$$

$$\frac{1}{2} \int u^{-\frac{1}{2}} du$$

$$\frac{1}{2} \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + C_3$$

$$\sqrt{u} + C_3$$

$$\sqrt{x^2-2} + C_3$$

$$\Rightarrow \sin^{-1}\left(\frac{x}{\sqrt{2}}\right) + \frac{1}{\sqrt{2}} \sec^{-1}\left(\frac{x}{\sqrt{2}}\right) + \sqrt{x^2-2} + C$$

$$\int \left(\frac{1}{x^2+3} + \frac{x}{x^2+3} + \frac{x^2}{x^2+3} \right) dx$$

$$\int \frac{1}{x^2+3} dx + \int \frac{x}{x^2+3} dx + \int \frac{x^2}{x^2+3} dx$$

$$\frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{x}{\sqrt{3}}\right) + C_1$$

$$\frac{1}{2} \ln|x^2+3| + C_2$$

١) حدوده
حدوده

٢) السط ١٥ المقام

قائمة موزونة

$$\int \frac{x^2}{x^2+3} dx$$

$$\int \left[1 + \frac{-3}{x^2+3} \right] dx$$

$$x - 3 \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{x}{\sqrt{3}}\right) + C_3$$

①
②
③ غير
④

$$\begin{array}{r} 1 \\ \hline x^2 \\ - x^2 + 3 \\ \hline -3 \end{array}$$

$$\Rightarrow \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{x}{\sqrt{3}}\right) + \frac{1}{2} \ln|x^2+3| + x - \frac{3}{\sqrt{3}} \tan^{-1}\left(\frac{x}{\sqrt{3}}\right) + C$$

$$\int x^3 3^{7x^2} dx$$

Let $t = x^2$

$$dt = 2x dx$$

$$\frac{1}{2} dt = x dx$$

$$\int \underline{x} \cdot x^2 3^{7x^2} \underline{dx}$$

$$\frac{1}{2} \int t 3^{7t} dt$$

$$u = t$$

$$du = dt$$

$$dv = 3^{7t} dt$$

$$v = \frac{3^{7t}}{7 \ln 3}$$

$$\frac{1}{2} \left[\frac{1}{7 \ln 3} t 3^{7t} - \frac{1}{7 \ln 3} \int 3^{7t} dt \right]$$

$$\frac{1}{2} \left[\frac{1}{7 \ln 3} t 3^{7t} - \frac{1}{7 \ln 3} \cdot \frac{3^{7t}}{7 \ln 3} \right] + C$$

$$\frac{1}{2} \left[\frac{1}{7 \ln 3} x^2 3^{7x^2} - \frac{1}{7 \ln 3} \cdot \frac{3^{7x^2}}{7 \ln 3} \right] + C$$

$$\int \frac{\cos^2 x \sqrt{\sin x}}{\tan x} dx$$



لازم رفیجات

$$\int \frac{\cos^2 x \cdot \sin^{\frac{1}{2}} x}{\frac{\sin x}{\cos x}} dx$$

$$\int \frac{\cos^3 x \cdot \sin^{\frac{1}{2}} x}{\sin x} dx \rightarrow \sin^{\frac{1}{2}-1} x$$

$$\int \cos^3 x \cdot \sin^{-\frac{1}{2}} x dx$$

$$\int \cos^2 x \cdot \sin^{-\frac{1}{2}} x \cos x dx$$

$$\int (1 - \sin^2 x) \sin^{-\frac{1}{2}} x \cos x dx$$

Let $u = \sin x$
 $du = \cos x dx$

$$\int (1 - u^2) u^{-\frac{1}{2}} du$$

$$\int (u^{-\frac{1}{2}} - u^{\frac{3}{2}}) du$$

$$= \frac{u^{\frac{1}{2}}}{\frac{1}{2}} - \frac{u^{\frac{5}{2}}}{\frac{5}{2}} + C$$

$$= 2 (\sin x)^{\frac{1}{2}} - \frac{2}{5} (\sin x)^{\frac{5}{2}} + C$$

$$\int \cot^{\frac{1}{2}} x \sin^{\frac{7}{2}} x dx$$

لازم رفیجات

$$\int \cot^{\frac{1}{2}} x \cdot \frac{1}{\csc^{\frac{7}{2}} x} dx$$

$$\int \cot^{\frac{1}{2}} x (\csc x)^{-\frac{7}{2}} dx$$

← فریبی

→ زوجی

*فشلوا!
الرفیجات

$$\int \frac{\cos^{\frac{1}{2}} x}{\sin^{\frac{1}{2}} x} \cdot \sin^{\frac{7}{2}} x dx$$

$$\int \cos^{\frac{1}{2}} x \sin^{\frac{6}{2}} x dx$$

$$\int \cos^{\frac{1}{2}} x \sin^3 x dx$$

$$\int \cos^{\frac{1}{2}} x \sin^2 x \sin x dx$$

$$\int \cos^{\frac{1}{2}} x (1 - \cos^2 x) \sin x dx$$

Let $u = \cos x$

$$du = -\sin x dx \rightarrow -du = \sin x dx$$

$$- \int u^{\frac{1}{2}} (1 - u^2) du$$

$$- \int (u^{\frac{1}{2}} - u^{\frac{5}{2}}) du = - \left[\frac{2}{3} u^{\frac{3}{2}} - \frac{2}{7} u^{\frac{7}{2}} \right] + C$$

$$= - \left[\frac{2}{3} (\cos x)^{\frac{3}{2}} - \frac{2}{7} (\cos x)^{\frac{7}{2}} \right] + C$$

$$\int \frac{\tan^5 x}{\sin 2x} dx$$

$$\int \frac{\tan^5 x}{2 \sin x \cos x} dx$$

$$\frac{1}{2} \int \frac{\sin^5 x}{\cos^5 x \sin x \cos x} dx = \frac{1}{2} \int \frac{\sin^4 x}{\cos^6 x} dx$$

$$\frac{1}{2} \int \frac{\sin^4 x}{\cos^6 x} dx$$

* فُشِّلُوا الرُّفُيَّاتِ *

$$\frac{1}{2} \int \frac{\sin^4 x}{\cos^4 x \cdot \cos^2 x} dx$$

$$\frac{1}{2} \int \tan^4 x \cdot \sec^2 x dx$$

Let $u = \tan x$
 $du = \sec^2 x dx$

$$\frac{1}{2} \int u^4 du$$

$$\frac{1}{2} \frac{u^5}{5} + C$$

$$\frac{1}{10} (\tan x)^5 + C$$

$$\int \cosh x \sin x \, dx = I$$

$$u = \sin x \quad dv = \cosh x \, dx$$

$$du = \cos x \, dx \quad v = \sinh x$$

$$\sin x \sinh x - \int \sinh x \cos x \, dx$$

$$u = \cos x \quad dv = \sinh x \, dx$$

$$du = -\sin x \, dx \quad v = \cosh x$$

$$\cos x \cosh x + \int \cosh x \sin x \, dx$$

$$I$$

$$I = \sin x \sinh x - [\cos x \cosh x + I]$$

$$I = \sin x \sinh x - \cos x \cosh x - I$$

$$2I = \sin x \sinh x - \cos x \cosh x$$

$$I = \frac{1}{2} [\sin x \sinh x - \cos x \cosh x] + C$$



$$\int \sec^3 x \, dx = I$$

$$\int \sec x \cdot \sec^2 x \, dx$$

$$u = \sec x \quad dv = \sec^2 x \, dx$$

$$du = \sec x \tan x \, dx \quad v = \tan x$$

1) By parts

2) حول

3) اضرب (الولبي)

$$\sec x \tan x - \int \sec x \tan^2 x \, dx$$

$$\int \sec x (\sec^2 x - 1) \, dx$$

$$\int (\sec^3 x - \sec x) \, dx$$

$$I - \ln |\sec x + \tan x|$$

$$I = \sec x \tan x - [I - \ln |\sec x + \tan x|]$$

$$I = \sec x \tan x - I + \ln |\sec x + \tan x|$$

$$2I = \sec x \tan x + \ln |\sec x + \tan x|$$

$$I = \frac{1}{2} [\sec x \tan x + \ln |\sec x + \tan x|] + C$$



$$\int \sec x \tan^2 x \, dx = I$$

$$\int \sec x \tan x \tan x \, dx$$

$$u = \tan x \quad dv = \sec x \tan x \, dx$$
$$du = \sec^2 x \, dx \quad v = \sec x$$

- 1) By parts
- 2) حول
- 3) اضرب (الولبي)

$$\tan x \sec x - \int \sec x \sec^2 x \, dx$$

$$\int \sec x (1 + \tan^2 x) \, dx$$

$$\int (\sec x + \sec x \tan^2 x) \, dx$$

$$\ln|\sec x + \tan x| + I$$

$$I = \tan x \sec x - [\ln|\sec x + \tan x| + I]$$

$$I = \tan x \sec x - \ln|\sec x + \tan x| - I$$

$$2I = \tan x \sec x - \ln|\sec x + \tan x|$$

$$I = \frac{1}{2} [\tan x \sec x - \ln|\sec x + \tan x|] + C$$